

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have made substantial progress across a wide range of visual question answering benchmarks, spanning visual reasoning, document understanding, and multimodal dialogue, with gains evident across many base models, alignment architectures, and training corpora. Yet recent work shows that these same models trail behind on traditional image classification benchmarks that probe fine-grained visual knowledge. This paper makes three contributions toward closing that gap. First, we deliver a large-scale evaluation of recent VLMs on fine-grained classification benchmarks, establishing that fine-grained recognition is a distinct axis of visual ability that current VLM benchmarks do not capture. Second, through a systematic ablation study, we isolate which model components and training choices drive fine-grained performance: we show that a better LLM lifts all benchmark scores roughly equally, whereas a better vision encoder disproportionately improves fine-grained classification. Third, we identify the pre-training stage as similarly decisive for fine-grained performance, especially when the language model weights are unfrozen during pretraining. Together, these contributions offer concrete, actionable guidance for enhancing fine-grained visual understanding and vision-centric capabilities in future VLMs.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which integrate vision encoders with large language models (LLMs), have demonstrated increasingly sophisticated capabilities in interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). These models can perform complex tasks such as open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite these advances, one crucial question remains open and motivates this paper: How well do VLMs perform on fine-grained visual perception tasks, and—just as importantly—what can we change to improve them? This paper answers both parts of that question, first by measuring where VLMs stand on fine-grained classification and then by pinpointing the design choices that move the needle.

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires models to focus on subtle distinguishing features while maintaining robustness to intra-class variations in pose, lighting, and background. While traditional vision encoders like CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have demonstrated strong performance on fine-grained classification benchmarks, the performance of newer VLMs on this axis has received little attention—a gap that our first contribution is designed to fill.

Why does this gap matter enough to study? Understanding the fine-grained knowledge capabilities of VLMs is essential because perception serves as the foundation for advanced capabilities such as understanding and reasoning. Real-world applications of VLMs often involve fine-grained, long-tailed distributions where distinguishing between closely related categories is crucial. For instance, if a model fails to identify a mushroom species correctly, it cannot determine whether it is poisonous based on textual knowledge or whether it is safe for consumption. Beyond such high-stakes use cases, insights into the factors influencing fine-grained knowledge acquisition can directly inform more effective architectures and training strategies for future VLMs—which is precisely what our second contribution provides.

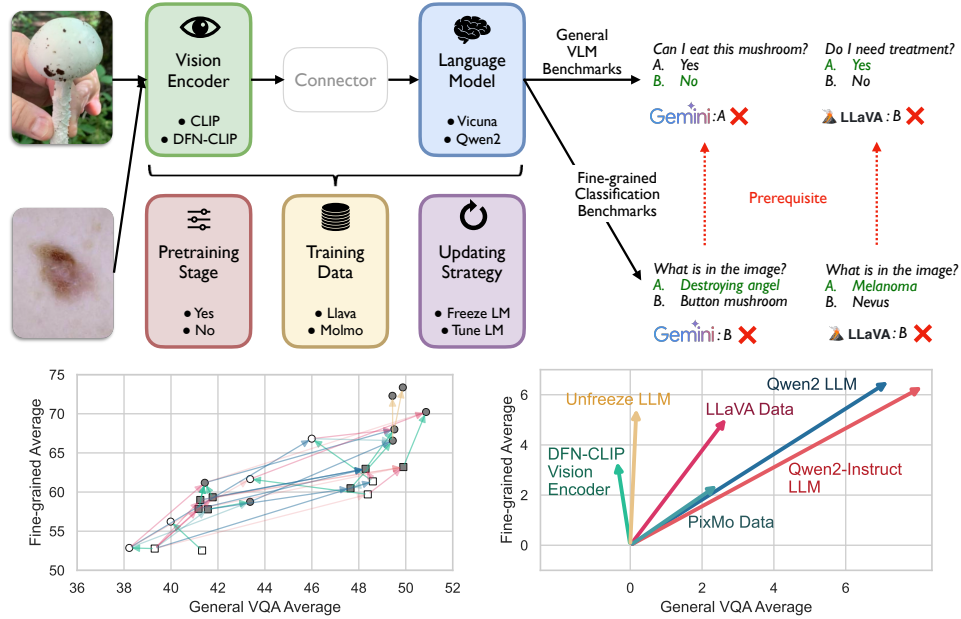


Figure 1: **Overview of our two contributions.** (Top) Contribution 1—we investigate the fine-grained classification capabilities of vision-language models (VLMs), a crucial yet often overlooked aspect that underpins higher-level understanding and reasoning. (Bottom) Contribution 2—through 22 systematic ablation experiments on key model components and training strategies (left), we identify the factors that drive fine-grained classification performance in VLMs (right).

Contribution 1: A comprehensive evaluation establishing fine-grained recognition as a distinct, unmeasured axis. We first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, comparing their performance on fine-grained classification benchmarks with their capabilities on general VLM tasks. Specifically, we evaluate 15 VLMs, such as LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks, including ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), alongside aggregated results from eight general VLM benchmarks such as MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). This evaluation yields two takeaways. First, fine-grained classification performance varies substantially among models with comparable general abilities, establishing that fine-grained recognition is a distinct axis of visual intelligence that existing VLM benchmarks fail to capture. Second, we document a persistent gap between VLMs and their own CLIP-based vision encoders, corroborating and broadening previous findings (Zhang et al., 2024; Geigle et al., 2024).

Contribution 2: A systematic ablation study identifying what drives fine-grained performance.

To move from diagnosis to actionable guidance, we conduct a systematic study of how key model components and training paradigms shape fine-grained classification. Concretely, we ask whether VLM components—such as language models and vision encoders—affect classification performance, and whether training strategies—such as pretraining or variations in data quality—influence results. Through 22 carefully designed ablation experiments, we isolate several factors that significantly affect fine-grained knowledge acquisition in VLMs. We summarize these factors below, each of which we substantiate in the referenced section:

- **Language model (§4.2):** Improvements in the base language model enhance performance uniformly across both fine-grained classification and general VLM benchmarks.
- **Vision encoder (§4.2):** A stronger vision encoder improves fine-grained classification performance but has a limited impact on general VLM benchmarks, particularly in models trained in a two-stage fashion.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning datasets significantly boosts fine-grained classification performance but has a lesser effect on general VLM benchmarks.

- **Weight updating method (§4.3):** Fine-grained classification performance improves when both the connector between the vision encoder and LLM, as well as the LLM itself, are trained during pretraining, compared to training only the connector.
- **Data quality (§4.3):** The quality of pretraining and instruction-tuning data has a limited impact on fine-grained classification, particularly when language model weights are frozen during pretraining.

In summary, this paper contributes (i) a comprehensive evaluation that surfaces fine-grained visual perception as an essential yet overlooked axis of current VLM benchmarks, and (ii) a systematic ablation study that identifies the specific components and training choices governing fine-grained knowledge acquisition. Taken together, these contributions offer concrete, reusable guidance for developing vision-centric VLMs with enhanced visual understanding, improving their effectiveness in real-world long-tail tasks that demand fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) are a class of models that integrate visual inputs with language models, typically following the LLaVA-based architecture (Liu et al., 2023) of three core components: a vision encoder, a language model, and an MLP connector bridging the two. These models have achieved strong performance across various multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). To evaluate VLMs, numerous benchmarks have been developed, such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). However, these benchmarks primarily assess reasoning and language understanding given visual inputs, while overlooking core vision-centric capabilities such as object recognition and fine-grained classification. It is exactly this blind spot that our first contribution targets: we systematically evaluate and analyze VLMs on fine-grained classification, an essential yet underexplored aspect of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that aims to distinguish between visually similar subcategories within broader object categories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). These benchmarks have played a key role in training and evaluating vision encoders, like ResNet (He et al., 2016), CLIP (Radford et al., 2021) and DINO (Caron et al., 2021). Despite their importance in vision research, recent VLMs have largely overlooked these datasets in their evaluation protocols in favor of visual question-answering. Because fine-grained classification is critical for real-world applications that involve long-tail distributions and require precise recognition (e.g., medical diagnosis, food safety, and species identification), evaluating VLMs on such benchmarks is essential for understanding their practical usability from a vision-centric perspective, beyond language-centric reasoning tasks—the motivation underlying our evaluation contribution.

Evaluation and ablation of VLMs for fine-grained classification. Our two contributions extend two distinct lines of prior work. For evaluation, while recent studies have explored VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), they have been limited in scope, often focusing on just a few models or datasets; our first contribution broadens this to a larger set of VLMs across multiple fine-grained classification benchmarks in a unified multiple-choice format. For analysis, our second contribution goes beyond this prior work with a comprehensive ablation study that pinpoints the key factors driving fine-grained classification performance.

This ablation contribution also connects to a separate strand of research on the VLM design space, which has analyzed aspects such as model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). Those studies, however, primarily target general benchmarks that are more language- and reasoning-focused. By contrast, we examine how key design choices—base model selection, pretraining methods, and the composition of pretraining and instruction-tuning data—affect fine-grained classification specifically, contributing novel insights into optimizing VLMs for vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section develops our first contribution: a comprehensive evaluation of fine-grained classification in existing VLMs. We motivate why this axis deserves attention, describe our benchmarks and evaluation setup, and then present the two central findings that emerge from testing 15 existing VLMs—(i) fine-grained classification is a distinct capability that current benchmarks miss, and (ii) VLMs lag well behind their own vision encoders.

3.1 MOTIVATION

Fine-grained classification is a fundamental task in computer vision that focuses on distinguishing visually similar subcategories within a broader category (e.g., different bird species or car models) (Wei et al., 2021). Many evaluation benchmarks, such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have been developed and widely used as standard testbeds for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) achieved near- or superhuman performance on challenging fine-grained classification tasks.

Despite this maturity, these benchmarks have been largely absent from the evaluation of vision-language models (VLMs)—the omission our evaluation contribution sets out to correct. While recent VLMs like LLaVA (Liu et al., 2023) have demonstrated strong capabilities in interpreting visual concepts, their evaluation is typically limited to visual question answering (VQA), such as academic VQA benchmarks like MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks like DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks emphasize reasoning and language processing but lack fine-grained object categories, thereby leaving core vision-centric capabilities largely unexamined.

This omission matters because fine-grained classification is crucial for VLMs: real-world applications often encounter long-tailed distributions that require precise visual knowledge to separate similar categories. For example, in daily life, if a blind person asks whether a mushroom is safe to eat, the model must first correctly identify the species before reasoning about its toxicity—misclassifying a deadly destroying angel as an edible white button mushroom could have fatal consequences. Similarly, in medical diagnosis, failing to distinguish between similar diseases could lead to incorrect treatment and serious health risks. In self-driving cars, confusing a “stop” sign with a “do not enter” sign—both red with similar shapes—could result in dangerous navigation errors. These stakes are exactly why quantifying, and then improving, VLMs’ fine-grained classification is the central concern of this paper.

3.2 BENCHMARKS

To support this evaluation contribution, we assess the fine-grained visual knowledge of VLMs using four well-established object recognition benchmarks, each covering a different domain of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks contain 37 to 1,000 distinct classes, which makes evaluating VLMs directly in the multiple-choice format—commonly used in VLM training and evaluation—challenging. To address this, we adopt the methodology of Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. Specifically, for each image, we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to select the four hardest negative choices based on image-label cosine similarity and combine them with the ground-truth label to form the answer set. We deliberately choose the OpenCLIP model to avoid biasing the difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023) (all trained on different data than the OpenCLIP model). Crucially, this conversion lets us test VLMs in their native format while preserving most of the difficulty: for instance, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the 102-class Flowers-102, rising only to 78.3% on our 5-way multiple-choice version.

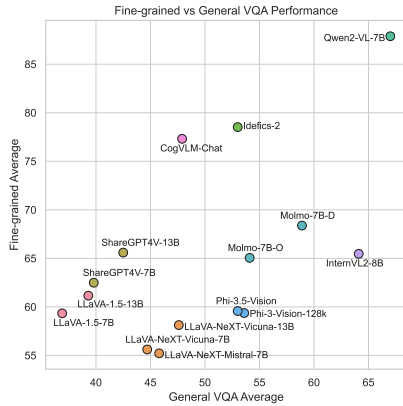


Figure 2: **Evidence for Finding 1: fine-grained classification versus general VQA performance across VLMs.** Recent VLMs with comparable general VQA scores differ widely in fine-grained accuracy, showing that fine-grained classification is a distinct axis of visual capability that standard VQA benchmarks fail to measure.

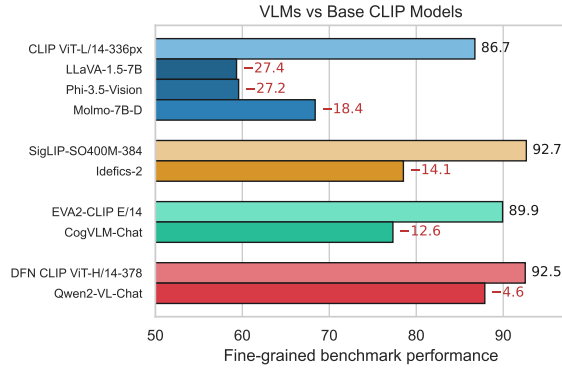


Figure 3: **Evidence for Finding 2: VLMs versus their own CLIP vision encoders in fine-grained classification.** While Qwen2-VL-Chat nearly matches its DFN-CLIP encoder, every other VLM falls well behind its encoder, quantifying how much fine-grained visual knowledge is lost in current VLMs and how much room remains for improvement.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use the respective default prompt format from VLMEvalKit (Duan et al., 2024) and measure accuracy as the exact match with the multiple-choice options. To ground these numbers against a strong reference point, we also evaluate various CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, where both images and class labels (“A photo of a [class]”) are encoded using CLIP’s image and text encoders, and the top-1 prediction is the label with maximum image–text cosine similarity.

The key comparison behind this contribution asks whether fine-grained ability is already captured by existing VLM benchmarks. To answer it, we place each VLM’s fine-grained classification score alongside its average over eight general VQA benchmarks, such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), from VLMEvalKit.

3.4 RESULTS

Our first result concerns whether existing benchmarks already measure fine-grained ability. Figure 2 plots fine-grained classification performance (averaged across our four benchmarks) against general VQA performance (averaged across eight existing VQA benchmarks) for all tested VLMs. The takeaway is clear: models with similar general VQA performance vary substantially in fine-grained classification accuracy. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both reach nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracy differs by 19pp (percentage points)—77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. Because fine-grained ability can swing this widely at a fixed general-VQA level, existing VLM benchmarks demonstrably do not capture it, which establishes fine-grained classification as a distinct dimension of visual intelligence and grounds our first contribution.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Our second result isolates where this lost ability comes from. Because every current VLM is built on an existing pretrained vision encoder, the multiple-choice format lets us directly compare a VLM against the very encoder it was built on. Figure 3 makes this comparison for selected VLMs and reveals a substantial gap in favor of the encoders. For instance, Molmo (Deitke et al., 2024), a very recent model that excels on general VQA benchmarks, obtains only 68.4% average accuracy in our multiple-choice setting, while its own CLIP ViT-L/14 (Radford et al., 2021) vision encoder reaches 86.7%. Even the strongest model we test, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points. That VLMs discard fine-grained knowledge already present in their encoders is precisely the opening our second contribution exploits: it motivates the ablation study of the next section, which pinpoints the design choices that recover this lost capability.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

This section develops our second contribution. Building on the observational results of Section 3, we ablate the key differences between models that may drive fine-grained classification, concentrating on base model selection and training methodology so that each factor can be attributed cleanly. All ablations build on the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). Every model undergoes one epoch of pretraining (when applicable) followed by one epoch of finetuning, and we hold LLaVA-1.5’s default hyperparameters—including learning rate and batch size—fixed across all experiments so that measured differences reflect the ablated factor rather than tuning.

Throughout, fine-grained performance is the average score across the four classification benchmarks of Section 3.2. For general VQA performance, we report the average over three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b), chosen for their domain coverage and widespread adoption in VLM evaluation.

4.2 BASE MODELS

Our analysis in Section 3 indicates that recent models typically adopt newer base LLMs, yet most still rely on the CLIP ViT-L/14 vision encoder from Radford et al. (2021). This asymmetry raises a natural question, which we answer here as the first part of our ablation contribution: how do the two base-model choices—LLM and vision encoder—separately affect fine-grained classification?

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we ablate the LLM component across different vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. The central result of this ablation is that a stronger LLM helps both axes roughly equally. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B improves performance across all configurations, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks; the instruction-tuned Qwen2-7B-Instruct behaves similarly (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base of Vicuna, yields only modest improvements, though the downstream implications of forgoing instruction tuning remain unclear. Importantly, all of our trained models—including the lower-performing Vicuna-based VLMs—consistently produce correctly formatted multiple-choice responses, so these fine-grained gains re-

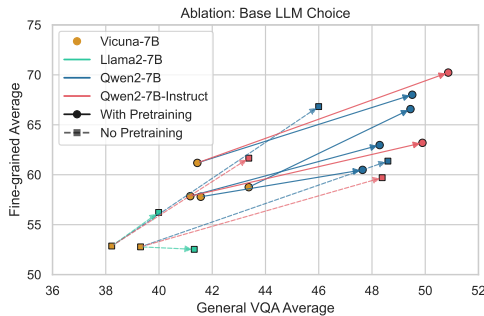


Figure 4: **LLM ablation (supports Takeaway 1): swapping Vicuna-7B in LLaVA for other LLMs.** A stronger LLM helps both axes about equally—switching from Vicuna to Qwen2-7B yields, on average, a +7.5pp gain in fine-grained performance and a +7.7pp gain in general VQA.

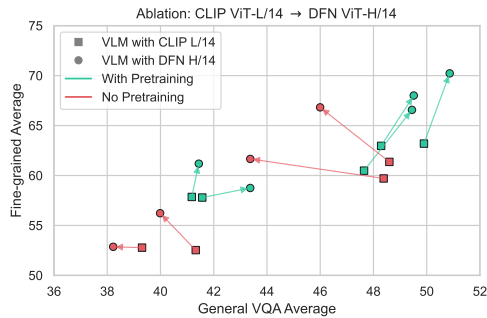


Figure 5: **Vision-encoder ablation (supports Takeaway 2): swapping CLIP ViT-L/14 in LLaVA for DFN-CLIP ViT-H/14.** A stronger encoder helps fine-grained ability disproportionately—given enough pretraining, switching to DFN-CLIP improves fine-grained performance by +4.5pp while moving general performance only +1.2pp.

flect improved knowledge about the presented options rather than better format adherence. This balanced improvement is what we distill into our first takeaway.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. However, Qwen2-VL (Wang et al., 2024b) uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which demonstrates superior zero-shot performance on object recognition tasks like ImageNet. We evaluate both vision encoders across various LLM choices and pretraining configurations.

Findings. In contrast to the LLM, a stronger vision encoder benefits fine-grained ability disproportionately—but only when it is properly integrated. Figure 5 shows that the effect of substituting CLIP L/14 with DFN-CLIP H/14 hinges on whether connector pretraining precedes finetuning. Without pretraining, DFN-CLIP even degrades general VQA performance (-2.5pp) while only modestly improving fine-grained scores (+2.8pp). Once connector pretraining is applied, however, general VQA performance holds steady or edges up (+1.2pp) while fine-grained performance rises substantially (+4.5pp). The lesson—captured in our second takeaway—is that a better encoder is a targeted lever for fine-grained ability, provided the encoder is first aligned through pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Having examined base-model choices, we now turn to the second part of our ablation contribution: the training pipeline. Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, trains on large-scale image-caption datasets with a captioning objective to align the vision encoder and LLM. The second stage, instruction tuning, refines the VLM with carefully curated multi-round, multi-modal instruction completion data, aligning it with human intent for real-world applications. We quantify the contribution of each stage to fine-grained ability in turn, beginning with pretraining.

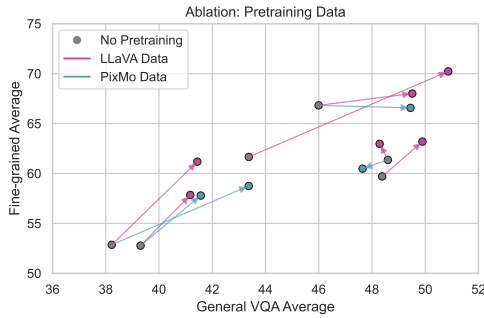


Figure 6: **Pretraining-data ablation (supports Takeaways 3–4).** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data gives similar gains, showing that the pretraining stage matters more than caption quality: LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp, and PixMo data by +2.4pp and +2.5pp.

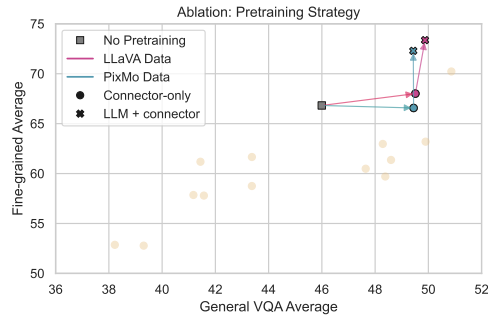


Figure 7: **Training-method ablation (supports Takeaway 5): connector-only pretraining versus LLM+connector tuning.** Unfreezing the LLM during pretraining is a further fine-grained lever, adding +5.5pp on fine-grained performance without hurting general VQA scores.

Table 1: **Summary of our ablation contribution: key settings versus existing models (bolded).** The table consolidates our takeaways: upgrading the base LLM and vision encoder and pretraining both the connector and LLM on sufficient data each benefit fine-grained performance, while instruction finetuning contributes least. General VQA follows a different pattern, where the choice of LLM plays the largest part.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

4.3.1 PRETRAINING

Setup. This ablation is organized around three questions about pretraining, each yielding a takeaway below:

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How important is the quality of pretraining data for downstream performance?

To answer them, we experiment with three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, with high-quality human-annotated detailed captions), and no pretraining. For most experiments, we pretrain only the connector following Liu et al. (2023). We additionally explore a strategy similar to Deitke et al. (2024), pretraining the connector for 20% of the steps and then unfreezing the LLM for the remaining 80%. All configurations are followed by one epoch of finetuning on LLaVA finetuning data.

Findings. Answering the first question, Figure 6 shows that connector-only pretraining generally improves VLM performance, with the exception of the Qwen2 and CLIP ViT-L/14 combination. The gains are largest for Vicuna-based VLMs, which improve on average by +6.1pp in fine-grained

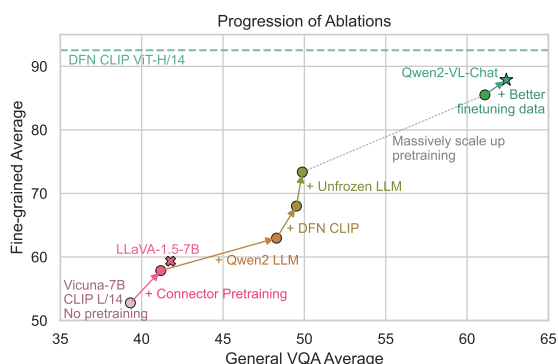


Figure 8: **Combined effect of our ablation findings.** Applying the levers identified above in sequence—base LLM, vision encoder, and training settings, starting from Vicuna-7B and CLIP ViT-L/14—raises fine-grained classification accuracy from 52.8% to 73.4%, closing more than half the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining gap primarily to Qwen2-VL’s far larger pretraining set of 1.4T tokens.

classification and +3.1pp in general VQA. As with the vision encoder, the effect is skewed toward fine-grained ability, giving our third takeaway.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Answering the third question, we find—surprisingly—almost no difference between pretraining on low-quality web-scraped captions (LLaVA) and highly detailed human annotations (PixMo): relative to LLaVA data, PixMo data shifts fine-grained performance by only -1.6pp and general VQA by +0.4pp. What matters is that pretraining happens, not how rich its captions are.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

Answering the second question, we hypothesize that connector-only pretraining may prevent the model from fully exploiting higher-quality captions. Testing this, we pretrain both the LLM and connector with a 20% connector-only warmup. Figure 7 shows that unfreezing the LLM markedly improves fine-grained classification (+5.5pp) without compromising general VQA scores. Notably, this gain appears with both LLaVA and PixMo data, indicating that training on the lower-quality captions does not impair the LLM’s language capabilities—and yielding our fifth takeaway.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. The final training-stage question is how much instruction finetuning contributes. Departing from our previous setup, we begin with the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (from which Qwen2-VL-7B-Chat is trained) and finetune on LLaVA-Instruct data (Liu et al., 2023) following our earlier protocol. Comparing against Qwen2-VL-7B-Chat, which uses 2M in-house instruction-finetuning examples (Wang et al., 2024b), isolates the effect of the finetuning dataset.

Findings. Finetuning solely on LLaVA data slightly reduces both axes: fine-grained scores fall 2.4pp (87.9% to 85.5%) and general benchmark scores fall 1.3pp (62.4% to 61.1%). As Figure 8 makes visible, this is the smallest fine-grained contribution among all our ablations, indicating that instruction finetuning is a far weaker lever for fine-grained knowledge than base-model selection or pretraining strategy—our sixth takeaway.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

We now consolidate the ablation contribution into a single picture. Figure 8 and Table 1 synthesize the six takeaways above, showing how each component contributes to closing the gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Beginning from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021)), applying our identified levers in sequence raises fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The consolidated view sharpens the division of labor among these levers: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately boost fine-grained ability, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) drives most of the general-VQA gain.

Our contribution also delimits what the ablations do *not* explain: a notable 12-point fine-grained gap to Qwen2-VL remains (Table 1). We weigh two candidate explanations—architectural differences and pretraining data scale—and conclude in favor of scale. While architecture could play some role, our observational results (Section 3.4) and training ablations (Section 4.3) point to pretraining data scale as the dominant factor. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) experiments used relatively small datasets—each fewer than 1M image-caption pairs, or roughly 200M and 400M tokens, respectively—whereas Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, orders of magnitude more. This disparity most plausibly accounts for the residual gap, reinforcing our broader message that extensive pretraining is central to developing VLMs with strong fine-grained classification.

5 LIMITATIONS & CONCLUSION

Two practical limitations bound the scope of our contributions and point to future work. First, due to computational constraints, our ablations compare training on <1M data points rather than the billion (B)-scale corpora used by some newer VLMs, so how large-scale training reshapes our conclusions remains open—indeed, this is the very factor we identify as the most likely explanation for the residual gap to Qwen2-VL. Second, newer work proposes alternative pretraining strategies that may interact differently with fine-grained visual understanding than the ones we ablate.

To conclude, this paper delivers two contributions. Our first contribution is a comprehensive evaluation that establishes fine-grained visual classification as a crucial yet underexplored dimension of VLMs—one that current benchmarks fail to measure and on which VLMs trail their own vision encoders. Our second contribution is a systematic ablation of key model components and training paradigms that identifies which choices actually improve fine-grained classification, from stronger vision encoders to unfreezing the LLM during pretraining. Together, these contributions turn a diagnosed weakness into actionable guidance for building more vision-centric VLMs, strengthening their applicability in real-world scenarios that demand precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed per-setting results supporting our ablation contribution (aggregated).** Full breakdown of every ablated configuration behind the takeaways summarized in the main text.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed ablation results on general VQA benchmarks.** Per-benchmark scores underlying the general-VQA column of our contribution summary.

A FULL RESULTS

For completeness, and to let readers verify the ablation contribution at the level of individual settings, we report the full results here: aggregated performance in Table 2, general VQA benchmarks in Table 3, and fine-grained classification in Table 4.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed ablation results on fine-grained classification benchmarks.** Per-benchmark scores underlying the fine-grained column of our contribution summary.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

To document the evaluation protocol behind our first contribution, we detail here the four traditional classification benchmarks on which we evaluate VLMs and base vision encoders, as introduced in Section 3.2:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset covering a broad range of supercategories, each containing multiple fine-grained subcategories, based on the WordNet hierarchy. The ImageNet test set contains 50,000 images with classification across 1,000 classes.

One subtlety deserves note: because of the ImageNet/WordNet class hierarchy, many classes are too visually distinct from the others to yield hard multiple-choice distractors (for instance, only three classes fall under the “person” category). To handle this, and to avoid the dataset’s excessive size, we use the human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting from 19,056 annotated examples, we narrow the pool to 5,457 images spanning 928 of the original 1,000 classes. Because this curation leaves the remaining classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species characterized by high intra-class variation and inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories with challenging variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories, exhibiting substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For every fine-grained classification test set except ImageNet (handled above), we generate hard negatives by running zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top 4 incorrect predictions and shuffle them with the correct label to form the answer choices.

Prompt formatting. In our multiple-choice VQA format, the question is always “What type of object is in this photo?”, and the answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompts and formats, we adopt each model’s corresponding format from VLMEvalKit (Duan et al., 2024)—for instance, “Answer with the option’s letter from the given choices directly.”

For CLIP models we use the standard zero-shot procedure, which has no question format; the classnames passed to the text encoder are instead formatted with the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational testing behind our first contribution (Section 3), we rely directly on the numbers reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCR-Bench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). We use this reported average only in Figure 2.

For all further evaluations—both on existing models and on the models we train for the ablation contribution in Section 4—we narrow to a subset of multiple-choice VLM benchmarks that permits a more direct, controlled comparison. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples testing a wide range of VLM capabilities split between perception and reasoning. While the majority of the examples deal with coarse-grained perception and visual reasoning tasks, some examples test “fine-grained perception”, though the categories (such as action recognition, attribute recognition, and OCR) are in a different domain from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a wide distribution of domains and applications. Though the test set has 10,500 examples, most works cite model performance on the validation split, which contains 900 questions. MMMU tests general knowledge tied to visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A dataset comprised of 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, both across domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). All existing models come with a predefined evaluation procedure; for our own trained models, we apply the LLaVA evaluation code, since we finetune exclusively on LLaVA-Instruct data.

C TRAINING

To make the ablation contribution reproducible, we document the full training setup here. We build on the LLaVA-1.5 training codebase (Liu et al., 2023) and, for the majority of experiments, retain the LLaVA architecture and training procedure.

Pretraining. Pretraining is one epoch on either the LLaVA pretraining data (CC-3M images and captions) or the Molmo pretraining data (PixMo-Cap); for PixMo-Cap we use only the “caption” field, not the raw audio transcript. We use a batch size of 256. In most runs we tune only the randomly initialized MLP connector at a learning rate of 1e-3. When ablating the pretraining strategy (the basis of Takeaway 5), we first warm up the connector alone for 20% of steps, then unfreeze the LLM and tune it as well at a learning rate of 2e-5 for the remaining 80%.

Finetuning. Finetuning is always one epoch on the LLaVA-1.5 finetuning data—LLaVA-Instruct plus VQA training data from various datasets, enabling proper benchmarking. We hold the LLM learning rate at 2e-5 with a batch size of 128, and the vision encoder stays frozen throughout both phases.

For the Qwen2-VL-architecture ablation, we start from Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models on 4 nodes of 4 A100 (40GB) GPUs each. Training typically finishes within 20 hours, varying with the CLIP model size.

LLaVA vs. PixMo data. The main text contrasts the caption quality of LLaVA and PixMo pre-training data, which underlies Takeaway 4. LLaVA pretraining data comes from CC-3M, whose web-scraped captions are known for low quality and relevance. PixMo-Cap, by contrast, is built from human annotators recording a 60–90 second audio description of each image and combining it into a proper caption (Deitke et al., 2024). A surface analysis makes the gap concrete: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words—yet, as Takeaway 4 reports, this large quality difference barely moves fine-grained performance.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”